

Four frameworks, one event — and the main causality axiom becomes a theorem

Bridge Document | DA-1 / T-SNAP Update

ZP Companion | Initial: April 2026 | Current: 1.14, September 2026

This companion explains the ideas in plain language with diagrams and real-world examples. It is not the formal document — every claim here restates a result already proved in the corresponding technical document. Consult that document for the authoritative mathematics.

What Is ZP-E Doing?

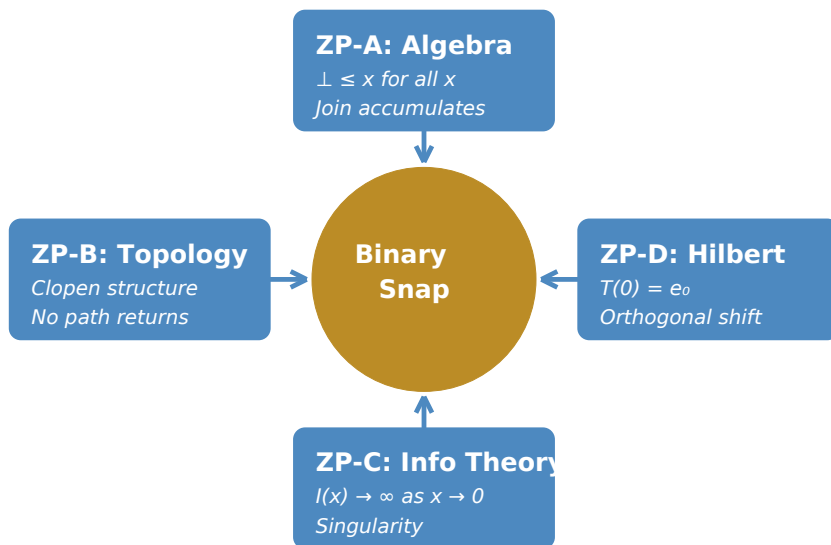
ZP-E is the cross-framework synthesis. Written last — after ZP-A through ZP-D are each internally closed — its job is to show that the four independent frameworks all describe the same event: the Binary Snap, from four different mathematical vantage points.

ZP-E does not re-derive anything. It imports closed results from each sub-document and shows their consistency. Where a cross-framework connection requires an assumption, that assumption is named explicitly as a bridge axiom — not hidden inside the argument.

The Four Descriptions of the Same Event

The Binary Snap — the transition from nothing ($\perp$) to the first state (ϵ_0) — looks different depending on which mathematical language you use. ZP-E's central result is that all four descriptions are consistent: one event, four angles.

ϵ_0 is the proof-theoretic ordinal of Peano Arithmetic, the minimum ordinal whose well-ordering PA cannot prove. Goodstein's theorem is the standard witness: every Goodstein sequence eventually terminates, but PA cannot prove this; the proof requires transfinite induction up to ϵ_0 . The same object has a second characterization as the minimum fixed point of the map $\alpha \mapsto \omega^\alpha$. The Binary Snap arrives at ϵ_0 via this fixed-point route, reaching the same object that the PA strength analysis identifies from the proof-theoretic side.



The Binary Snap (amber) described simultaneously in all four frameworks. Each arrow is an independent mathematical description of the same event.

The Binary Snap (amber center) described simultaneously in all four frameworks. Each arrow represents an independent mathematical description of the same event.

Real-world example — A car crash described by four witnesses

An engineer (forces), a doctor (injuries), a lawyer (liability), and a physicist (energy) each describe the same crash completely within their own discipline. ZP-E shows the four mathematical frameworks are in exactly this relationship to the Binary Snap.

Remember: The car crash illustrates what it means to describe one event in multiple frameworks. The Zero Paradox is not about car crashes — the car crash is an analogy for the first transition from $\perp$ to ϵ_0 — a transition the framework commits to, rather than one any join-semilattice compels. A join-semilattice in which T-SNAP holds and nothing moves is exhibited in the Lean source.

The Central Advance: AX-1's Shape Half is Now a Theorem

In earlier versions, the Binary Snap causality was listed as AX-1 — an axiom: a foundational assumption that could not be derived. The claim was: when P_0 is reached, the Snap happens. Why? Because AX-1 says so. That claim bundled two things: the shape of the Snap (what a step off $\perp$ looks like) and its occurrence (that the step is taken). T-SNAP now proves the shape half. The occurrence half was never retired: it is still a commitment.

The DA-1 insert changes this. The argument is now complete: reaching P_0 means a live machine configuration exists (DA-1). Any live configuration passes through c_1 (definition). c_1 is not $\perp$ (L-RUN). No program avoids this (TQ-IH). A state change from $\perp$ to any state other than $\perp$ is the Binary Snap (ZP-A D2). The Snap is derived, and that sentence has two true readings. The Lean proof is a fact about a fixed two-state machine type and takes no assumptions: it fixes the shape of the step, not that the step happens. The argument about the framework's own sequence uses commitments, among them AX-B1 (existence is binary):

a state exists or it does not) and CC-1 (the sequence starts at $\perp$); that the step actually happens rests further on DA-1, which is closed only given DP-2, and on the framework's commitment that instantiation occurs. The formal ZP-E states these premises under Premises of T-SNAP, and the Lean `t_snap_given` takes two of them as its hypotheses: the start at $\perp$ and the first step being taken.

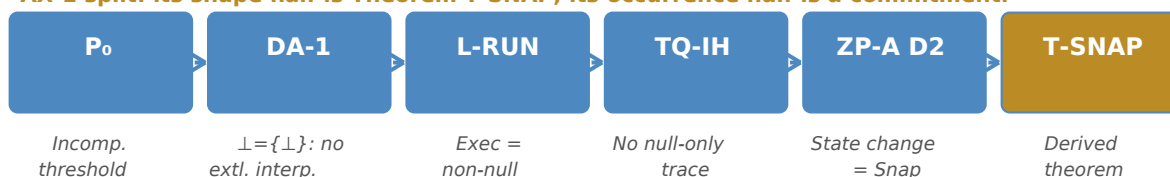
DA-1 is now a Derived Proposition rather than a freestanding Design Principle. Previously DA-1 was an honest but freestanding commitment: "a configuration at P_0 is necessarily executing." Now it follows from ZP-A CC-2, $\perp = \{\perp\}$, which is itself a commitment (argued, not proved). The bottom element $\perp$ is a Quine atom — a self-containing object with no external position from which it could be interpreted as a static description. A thing that interprets itself cannot be waiting for an external interpreter. So $\perp$ at P_0 is necessarily executing. The design commitment has become a derivation from that commitment.

The two-layer structure of DA-1: DA-1 rests on two explicit layers. The first is the formal conditional: DP-2 (Execution Distinguishability) establishes that machine states carry execution history independently of output values. From DP-2, Lean 4 can derive `da1_minimal_path` — a proof that before and after instantiation produce the same output value ($c_0 \rightarrow c_1$). `#print axioms` confirms zero axiom dependencies. This is the first Lean formalization of the core DA-1 claim, not just the surrounding algebra. The second layer asks: does $\perp$ actually satisfy DP-2's precondition? That case rests on three converging arguments.

Three paths to the precondition: (1) CC-2/R3 (ZP-A): $\perp = \{\perp\}$ is a Quine atom — it interprets itself, leaving no external position from which it could be read as a static description (above). (2) L-INF (ZP-C): the surprisal of $\perp$ diverges to infinity — no finite static distribution can represent $\perp$. (3) AIT bridge: at the incompressibility threshold P_0 , the description of $\perp$ is maximally incompressible — $K(c_1|n)/|c_1| = 1$. A string that cannot be compressed beyond itself must be its own execution; a static-description reading is ruled out by information theory alone. All three share D7's static/executing dichotomy as background and none is circular with DP-2.

Lean 4 formal closure (ZP-K): The three paths are not only conceptually convincing — two of them are now machine-checked. ZP-K adds a KleeneStructure instance for MachinePhase: it provides a concrete computational Quine (a code that is its own program, via Kleene's second recursion theorem), and proves that this Quine and the AFA self-containment argument are the same structural fact in two different languages. The result is `da1_closed_concrete`: in Lean 4, `IsQuineAtom( $\perp$  : MachinePhase)` is a proved theorem. Path 1 (AFA self-execution) and Path 3 (computational Kleene fixed point) are now formally IN LEAN SCOPE. Path 2 (informational bridge — unbounded surprisal $\rightarrow$ necessarily executing) remains a structural claim outside current Lean formalization. The formal grounding of DA-1 is therefore: DP-2 plus two Lean-verified structural paths.

AX-1 split: its shape half is Theorem T-SNAP; its occurrence half is a commitment.



The T-SNAP derivation chain. The chain uses commitments, among them AX-B1 and CC-1 ($S_0 = \perp$), and fixes the shape of the Snap; that the Snap occurs rests further on DA-1 (closed given DP-2) and on the commitment that instantiation

occurs. The shape half of AX-1 is now a theorem (T-SNAP, amber); its occurrence half is that commitment.

Real-world example — A legal case proved in part

A charge can bundle two claims: "the defendant was at the scene" and "the defendant acted." Surveillance footage can prove the first while the second still rests on testimony the court chooses to accept. AX-1 bundled two claims in the same way. T-SNAP proves the first, the shape of the Snap, which was once assumed. The second, that the Snap happens, was not proved along with it and remains a commitment.

What the Framework Still Assumes

After T-SNAP, the framework states three named axioms:

Commitment	Statement
AX-B1	Binary Existence. A state either exists or it does not. No third option. The framework’s ONE substantive modelling commitment: that the outcome space is DISCRETE rather than a continuum. The two states being distinct is decidable; the choice of a discrete alphabet is not, and the real numbers are where it fails.
AX-G1	Initial Object Exists. There is a starting point that reaches everything. Not a novel commitment — grounded in $\perp$ as the bottom element of the ZP-A semilattice.
AX-G2	Source Asymmetry. Nothing returns to the initial object. The origin is unreachable from outside. Not a novel commitment — follows from ZP-A antisymmetry and ZP-B C3.

AX-1 is not on this list: its shape half is Theorem T-SNAP. Further commitments are not axioms and are stated where they are used, among them AX-1’s occurrence half (that the Snap happens), CC-1 ($S_0 = \perp$, a ZP-A Conditional Claim), the choice of starting point that T-SNAP is derived given, and CC-2 ($\perp = \{\perp\}$, a ZP-A Forced Metatheoretic Commitment), which DA-1 follows from. The framework makes no stronger claim than it has to.

Remember: The structural results — monotonicity, clopen separation, informational singularity, orthogonal shifts — hold in any instantiation of the framework. The framework makes no claims about which physical theory, if any, instantiates it.